

STOCHASTIC MODELING

MODULE 7 | LESSON 3

Scale-free Networks

Reading Time	45 minutes
---------------------	------------

Prior Knowledge	Random Networks, Erdos-Renyi model
------------------------	------------------------------------

|Keywords | Scale-free networks, preferential attachment, Barabasi-Albert model

In the previous lesson, we learned that **Erdős–Rényi (ER) random networks** provide a useful benchmark for understanding connectivity and randomness.

However, when researchers began mapping *real networks* — social, biological, technological, and financial — they noticed key discrepancies:

1. **Heterogeneous connectivity:**

- In real systems, a few nodes (called **hubs**) have *many more connections* than others.
- ER networks, in contrast, predict that most nodes have similar degrees.

2. **Heavy-tailed degree distributions:**

- Real networks exhibit degree distributions that decay slowly — often following a **power law**, not a Poisson form.

3. **Robust yet fragile behavior:**

- These networks are resilient to random failures (removing small nodes) but extremely vulnerable if hubs fail.

In finance, this structure is particularly relevant:

- A few **systemically important institutions** (major banks, funds, clearing houses) are linked to many others.
- Their central position makes them vital for liquidity and information flow — but also potential points of contagion.

This observation motivated the study of **scale-free networks**.

1. Scale-Free Networks

A **scale-free network** is characterized by a **power-law degree distribution**:

$$P(k) \sim k^{-\gamma}, \quad \text{typically } 2 < \gamma < 3$$

This means that:

- Most nodes have few links,
- But a small number of nodes (hubs) have a disproportionately large number of connections.

Unlike in random networks, there is **no typical scale** for the degree — hence the term *scale-free*.

Power-law degree distributions have been observed in many complex systems:

Domain	Example	Interpretation
Internet	Autonomous systems connected by routers	Few highly connected backbone nodes
Social	Followers in online networks	Influencers vs typical users
Biological	Protein interaction networks	Essential proteins interacting with many others
Financial	Interbank lending, stock correlations, fund holdings	Systemic hubs connecting large portions of the system

In all these cases, the same property emerges: A small fraction of nodes concentrates most of the connections, creating a **hub-dominated structure**.

This implies that the probability of observing a node with degree much larger than the mean (i.e., **hub**) remains significant even as the network grows.

As a result, **scale-free networks are highly heterogeneous**, fundamentally different from the homogeneous randomness of Erdős–Rényi graphs.

2. The Barabási–Albert (BA) Model

The **Barabási–Albert (BA)** model (1999) explains how *scale-free networks* can emerge naturally from two simple mechanisms:

1. Growth:

The network expands over time — new nodes keep joining.

2. Preferential attachment:

New nodes are more likely to connect to **already well-connected nodes**.

This reflects the principle of “*the rich get richer*” or *cumulative advantage*.**

2.1. Model definition

Start with a small seed network of m_0 nodes, fully connected.

At each time step t :

1. A new node v_t is added.
2. It forms $m \leq m_0$ links to existing nodes. (**growth property**)
3. The probability that it connects to an existing node i is proportional to its degree k_i (**preferential attachment**):

$$P_i(k_i) = \frac{k_i}{\sum_j k_j}$$

This means that highly connected nodes attract new connections more easily.

2.2 Expected Degree Growth

Let $k_i(t)$ be the degree of node i at time t . At each time step, node i 's expected degree increase is given by:

$$\frac{d k_i}{d t} = m \cdot P_i = m \cdot \frac{k_i}{\sum_j k_j}$$

Each new node adds m edges, contributing $2m$ degree units to the network. After t steps, the total degree would be:

$$\sum_j k_j = 2 m t$$

And, hence:

$$\frac{d k_i}{d t} = m \cdot \frac{k_i}{2 m t} = \frac{k_i}{2 t}$$

This means that the growth rate of a node's degree is:

- proportional to its current degree k_i (as in a 'richer-get-richer' paradigm), and
- inversely proportional to the current age t (growth slows as network gets bigger).

Solving for $k_i(t)$:

The next step is to obtain an expression for the degree of a given node i at a given time t . That is, the number of connections a node has at a given moment in time.

In order to derive $k_i(t)$, we start from the previous expression:

$$\frac{1}{k_i} \frac{d k_i}{d t} = \frac{1}{2 t}$$

When node i joins the network at time t_i , it has degree $k_i(t_i) = m$. At the current time t , its degree is $k_i(t)$. We can then integrate both sides over these intervals:

$$\int_m^{k_i(t)} \frac{d k}{k} = \frac{1}{2} \int_{t_i}^t \frac{d t'}{t'}$$

Both of these integrals yield logarithms:

$$\ln k_i(t) - \ln m = \frac{1}{2} (\ln t - \ln t_i)$$

$$\text{which simplify to: } \ln \frac{k_i(t)}{m} = \frac{1}{2} \ln \frac{t}{t_i}$$

and, using exponents on both sides:

$$\frac{k_i(t)}{m} = \left(\frac{t}{t_i} \right)^{1/2}$$

$$\Rightarrow k_i(t) = m \left(\frac{t}{t_i} \right)^{1/2}$$

As time increases, t / t_i grows, and so does $k_i(t)$. Nodes that joined earlier (smaller t_i) accumulate links faster, becoming **hubs**. This result shows that **degree grows as the square root of network age**, capturing the “**rich-get-richer**” dynamic that drives the formation of hubs in the Barabási–Albert model.

2.3. Degree distribution

So, given the growth dynamics of the network, **what's the probability that a chosen node has degree k ?**

To answer this question we need to know the degree distribution, $P(k)$.

From $k_i(t)$ to $P(k)$:

We already know that:

$$k_i(t) = m \left(\frac{t}{t_i} \right)^{1/2}$$

Hence,

$$t_i = m^2 t / k^2$$

Nodes are added **uniformly** over time, one per step, so “birth times” are uniformly distributed in the interval $[1, t]$:

$$P(t_i) = \frac{1}{t}$$

We can use a **change of variable** here, using the probability conservation equation:

$$P(k)dk = P(t_i)dt_i$$

Then, since we know $t_i = m^2 t / k^2$, we can easily get:

$$\frac{dt_i}{dk} = - \frac{2 m^2 t}{k^3}$$

Since t_i decreases as k increases, we take the absolute value:

$$\left| \frac{dt_i}{dk} \right| = \frac{2 m^2 t}{k^3}$$

Substitute this result into the probability conservation equation:

$$P(k) = P(t_i) \left| \frac{dt_i}{dk} \right|$$

Using $P(t_i) = \frac{1}{t}$ (nodes are added uniformly in time), we obtain:

$$P(k) = \frac{1}{t} \cdot \frac{2 m^2 t}{k^3} = \frac{2 m^2}{k^3}$$

Thus, the degree distribution follows a **power law**:

$$P(k) \propto k^{-3}$$

In a nutshell, the BA model offers some interesting features:

Property	
Degree distribution	$P(k) \sim k^{-3}$
Clustering coefficient	$C(k) \sim k^{-1}$ (high-degree nodes cluster less)
Average path length	$L \sim \frac{\ln n}{\ln \ln n}$ (very small)
Presence of hubs	A few nodes accumulate a large fraction of edges

These properties imply that BA networks are:

- **Highly connected**,
- **Strongly heterogeneous**,
- **Efficient** in communication,
- But **vulnerable** to targeted attacks on hubs.

There are many real-life settings where the "preferential attachment" feature of the BA model makes a lot of sense:

Example	Interpretation of "preferential attachment"
Interbank markets	Large banks attract more counterparties
Investment funds	Popular funds attract more investors over time
Corporate ownership	Major firms accumulate cross-holdings and partnerships
Financial contagion	Hubs become central transmission points of shocks

The BA model helps explain why **systemic importance emerges endogenously**, i.e., why big players naturally become more connected simply because they are already big.

3. The BA model in python

Now, the natural next step is to simulate the BA model and compare it to the random model of Erdős–Rényi that we already know. Let's do it!

```
In [1]: import networkx as nx
import matplotlib.pyplot as plt
```

```
import numpy as np
from collections import Counter
```

```
In [2]: # --- Parameters ---
n = 300      # number of nodes
m = 3        # new edges per node in BA model
p = (2*m) / (n-1) # ER connection probability for same expected avg degree
```

```
In [3]: # --- Generating networks ---
G_ba = nx.barabasi_albert_graph(n, m, seed=42)
G_er = nx.erdos_renyi_graph(n, p, seed=42)
```

Let's observe the differences visually first...

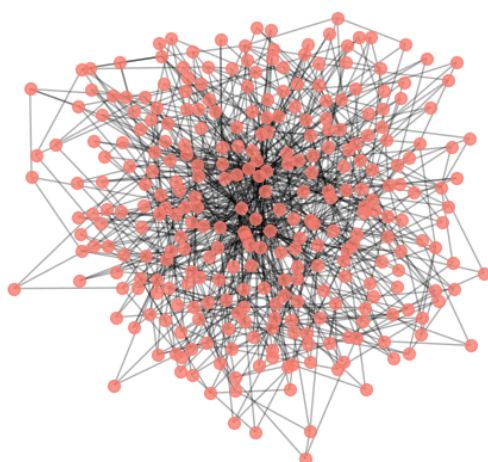
```
In [4]: plt.figure(figsize=(10, 5))

# --- Barabási-Albert (scale-free) network ---
plt.subplot(1, 2, 1)
pos_ba = nx.spring_layout(G_ba, seed=42) # force-directed layout for nice visual
nx.draw_networkx_nodes(G_ba, pos_ba, node_size=50, node_color="salmon", alpha=0.3)
nx.draw_networkx_edges(G_ba, pos_ba, alpha=0.3)
plt.title("Barabási-Albert (BA) Network — Scale-Free")
plt.axis("off")

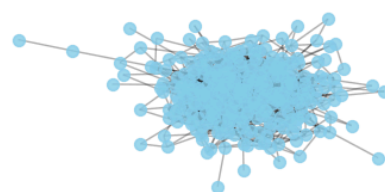
# --- Erdős-Rényi (random) network ---
plt.subplot(1, 2, 2)
pos_er = nx.spring_layout(G_er, seed=42)
nx.draw_networkx_nodes(G_er, pos_er, node_size=50, node_color="skyblue", alpha=0.3)
nx.draw_networkx_edges(G_er, pos_er, alpha=0.3)
plt.title("Erdős-Rényi (ER) Network — Random")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Barabási-Albert (BA) Network — Scale-Free



Erdős-Rényi (ER) Network — Random



As we probably expected, in the BA network, nodes gravitate visually toward a few high-degree hubs. In the ER network, on the other hand, there's no dominant node — connections look random and diffuse. Let's see this with more "hard data" by computing some metrics:

```
In [5]: # --- Compute basic metrics ---
def summarize_network(G, name):
    avg_deg = np.mean([d for _, d in G.degree()])
    clustering = nx.average_clustering(G)
    density = nx.density(G)
    print(f"{name} network")
    print(f"Nodes: {G.number_of_nodes()}, Edges: {G.number_of_edges()}")
    print(f"Average degree: {avg_deg:.2f}")
    print(f"Density: {density:.4f}")
    print(f"Average clustering coefficient: {clustering:.3f}\n")

summarize_network(G_ba, "Barabási-Albert (BA)")
summarize_network(G_er, "Erdős-Rényi (ER)")
```

Barabási-Albert (BA) network
Nodes: 300, Edges: 891
Average degree: 5.94
Density: 0.0199
Average clustering coefficient: 0.073

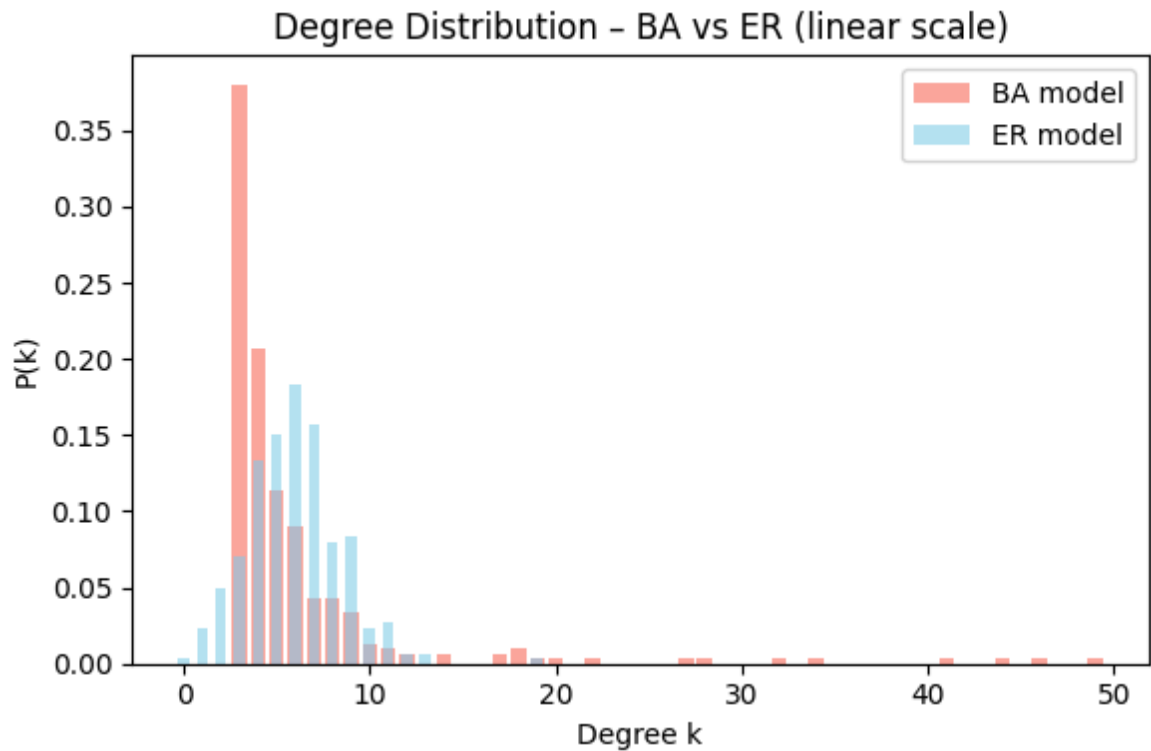
Erdős-Rényi (ER) network
Nodes: 300, Edges: 894
Average degree: 5.96
Density: 0.0199
Average clustering coefficient: 0.018

Also, in terms of degree distribution:

```
In [6]: # --- Degree distributions ---
def degree_distribution(G):
    degrees = [d for _, d in G.degree()]
    cnt = Counter(degrees)
    k, freq = zip(*sorted(cnt.items()))
    pk = np.array(freq) / sum(freq)
    return np.array(k), pk

k_ba, pk_ba = degree_distribution(G_ba)
k_er, pk_er = degree_distribution(G_er)
```

```
In [7]: # --- Plot ---
plt.figure(figsize=(6,4))
plt.bar(k_ba, pk_ba, width=0.8, alpha=0.7, label="BA model", color="salmon")
plt.bar(k_er, pk_er, width=0.6, alpha=0.6, label="ER model", color="skyblue")
plt.xlabel("Degree k")
plt.ylabel("P(k)")
plt.title("Degree Distribution - BA vs ER (linear scale)")
plt.legend()
plt.tight_layout()
plt.show()
```



Finally, we can verify the **power-law** in the BA model. That is, the probability of a node having degree k decays as a power of k :

$$P(k) \sim k^{-\gamma}$$

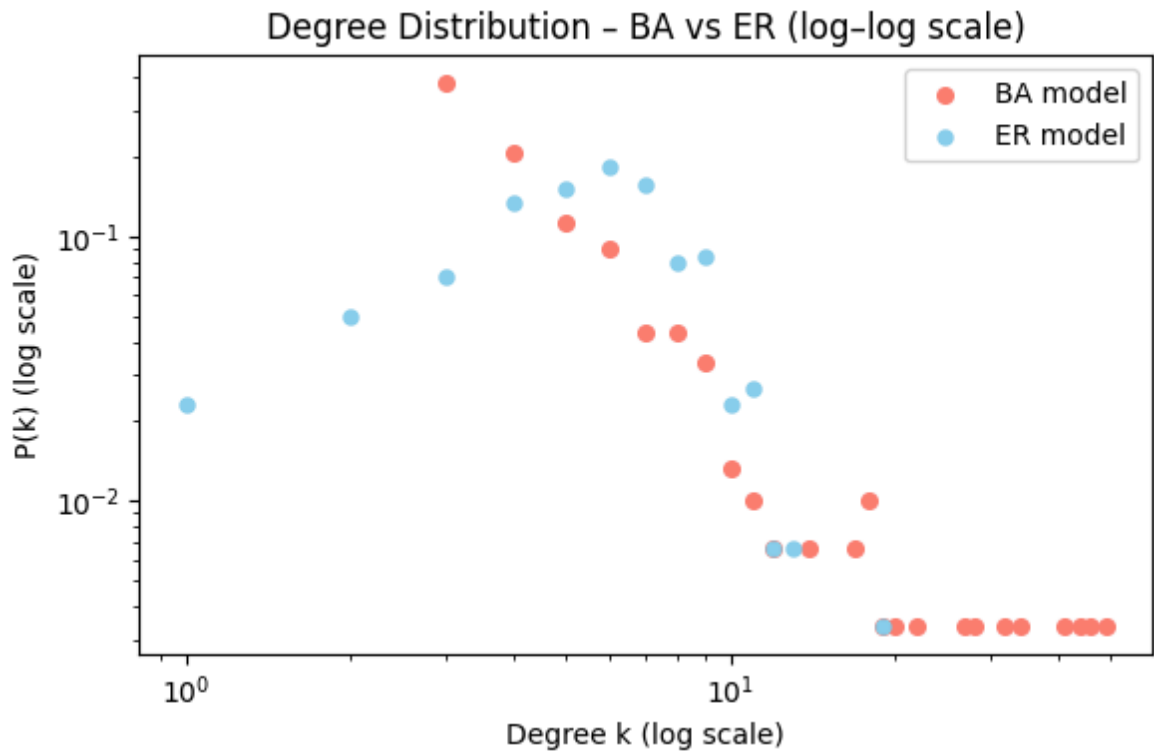
where γ is the **power-law exponent**.

Unlike the bell-shaped (Poisson) distribution of random networks (ER), the power-law form has a **heavy tail**:

- Most nodes have few links.
- A few nodes — the **hubs** — have extremely high connectivity.

Let's check this in our synthetic network:

```
In [8]: # --- Plot: Log-Log scale ---
plt.figure(figsize=(6,4))
plt.scatter(k_ba, pk_ba, s=30, label="BA model", color="salmon")
plt.scatter(k_er, pk_er, s=25, label="ER model", color="skyblue")
plt.xscale("log")
plt.yscale("log")
plt.xlabel("Degree k (log scale)")
plt.ylabel("P(k) (log scale)")
plt.title("Degree Distribution - BA vs ER (log-log scale)")
plt.legend()
plt.tight_layout()
plt.show()
```



4. Conclusion

Well done! In this lesson we have advanced our understanding from purely random network models to one that tries to explain why **hubs** appear in real-life networks as a function of growth and preferential attachment.

In the next lesson, we will see a real life implementation of such a model!

Copyright 2025 WorldQuant University. This content is licensed solely for personal use. Redistribution or publication of this material is strictly prohibited.